

ON THE CONSTRUCTION OF SEQUENTIAL SELECTION PROCEDURES

J.W.H. Swanepoel and J.H. Venter

Potchefstroom University

Key words: Selection procedures, sequential procedures, sequential elimination; largest mean; smallest variance; martingales.

SUMMARY

Contributions are made towards the problem of elimination sequential selection procedures as formulated by Hoel (AMS 42 p. 630); a general class of procedures is given and an alternative to Hoel's main result is derived enabling confirmation of his conjecture for the problem of selecting the stochastically largest population. A general method for calculating the asymptotic expected number of observations as the probability of correct selection tends to one is presented. Applications of these methods to the problems of selecting populations with largest mean or smallest variance are discussed; in each case a class of procedures is given and the best ones in the above asymptotic sense are found. Comparisons with existing procedures are discussed.

1. Introduction

Let $\Pi_1, \Pi_2, \dots, \Pi_k$ represent k populations ($k \geq 2$) and let $X_{i\alpha}$ denote the α -th observation on population Π_i , all such observations being assumed independent random variables. Further we assume that the distribution function of Π_i depends on some (unknown) parameter(s) θ_i and we write $\theta = (\theta_1, \dots, \theta_k)$ and let $\Theta = \{\theta\}$ denote the set of allowable parameter values. For each i and j , let $\tau_{ij} = \tau_{ij}(\theta)$ denote a given numerical measure of "preference" of population i over j ; τ_{ii} will also be supposed to be defined — usually we take $\tau_{ii} = 0$ or 1 depending on what is convenient under circumstances.

Following Hoel (1971) we define $\tau = \max_{i \neq j} \min_{j \neq i} \tau_{ij}$ and we let Π_k be a population for which $\tau = \min_{j \neq k} \tau_{kj}$, calling Π_k the "best" population. Let P^* and τ^* be given constants with $1 > P^* > 1/k$ and $\tau^* > \tau_{ij}$; for a given procedure for selecting Π_k , let CS denote the event of a correct selection; then we wish to construct procedures satisfying

$$(1) \quad P(\text{CS}) \geq P^* \text{ whenever } \tau \geq \tau^*.$$

A general procedure to this effect is given by Hoel (1971); here we show that his

method is a special case of a general class of procedures and we also give alternative conditions under which his method yields a procedure satisfying (1). Further, we give a general result by means of which the asymptotic expected number of observations (AENO) and asymptotic expected number of stages (AENS) as $P^* \rightarrow 1$ can be computed. In section 5.1 we apply these results to the construction of a class of procedures for the problem of selecting the population with largest mean and we show how a procedure having smallest AENO as $P^* \rightarrow 1$ within this class can be found. We then specialize to the normal case, finding that the "best" procedure is not that of Paulson (Hoel (1971), p. 633). In paragraph 5.2 we discuss procedures for selecting the normal population with smallest variance obtaining results analogous to the normal means case. Finally in paragraph 5.3 some other applications are indicated.

Notation: Throughout this paper we use the notation of Hoel (1971) as far as possible. In particular, we assume the existence of a non-increasing function h such that $\tau_{ji} = h(\tau_{ij})$ for all i and j and we write $\tau_0 = h(\tau^*)$. Also we remark that the "difference zone" $\Theta^* = \{\theta : \tau \geq \tau^*\}$ can be put in the equivalent form $\{\theta : \tau_{\kappa j} \geq \tau^* \text{ for all } j \neq \kappa\}$.

2. Procedures based on martingales

For each i and j , $i \neq j$, and n let

$$(2) \quad Z_{ijn} = f_n(X_{i1}, \dots, X_{in}; X_{j1}, \dots, X_{jn})$$

with f_n a given real-valued Borel-measurable function of the displayed arguments and define a sequential elimination procedure as follows: At stage one, take a single observation from each population and calculate the statistics Z_{ij1} ; if for some i , $Z_{ij1} \geq \epsilon = (k-1)/(1-P^*)$, then Π_j is eliminated from further consideration. If no populations remain after application of this rule, then we terminate and select as best a population Π_i for which $\min_{j \neq i} Z_{ij1}$ is maximized; if one population remains, it is selected as best; if more than one remain we proceed to the next (second) stage in which a single observation is made from each of the remaining populations, the statistics Z_{ij2} are calculated for all remaining pairs and Π_j is eliminated if for some remaining Π_i , $Z_{ij2} \geq \epsilon$. We continue in this manner until a population is selected.

Now suppose that for each $i \neq \kappa$ we can define random variables $Z_{i\kappa n}(\tau_{i\kappa})$ such that

(A) $\{Z_{i\kappa n}(\tau_{i\kappa}); n = 1, 2, \dots\}$ forms a non-negative martingale sequence;

(B) $E Z_{i\kappa n}(\tau_{i\kappa}) = 1$;

(C) $Z_{i\kappa n}(\tau_{i\kappa})$ is a non-increasing function of $\tau_{i\kappa}$ for $\tau_{i\kappa} \leq \tau_0 = h(\tau^*)$

with $Z_{i\kappa n}(\tau_0) = Z_{i\kappa n}$.

Then we have

Theorem 1:

If (A), (B) and (C) hold and the procedure stops with probability 1, then (1) holds.

Proof:

Since the procedure is assumed to stop with probability one, we get

$$1 - P(\text{CS}) \leq \sum_{i \neq k} P(\Pi_i \text{ eliminates } \Pi_k) \\ \leq \sum_{i \neq k} P(Z_{ikn}(\tau_o) \geq \epsilon \text{ for some } n \geq 1).$$

Now over Θ^* $\tau_{ki} \geq \tau^*$; hence $\tau_{ik} = h(\tau_{ki}) \leq h(\tau^*) = \tau_o$; since $Z_{ikn}(\tau_o) \geq \epsilon$ for some n implies $Z_{ikn}(\tau_{ik}) \geq \epsilon$ for some n by (C), we get

$$1 - P(\text{CS}) \leq \sum_{i \neq k} P(Z_{ikn}(\tau_{ik}) \geq \epsilon \text{ for some } n \geq 1).$$

By (A) and (B) and the well-known martingale inequality we get

$$1 - P(\text{CS}) \leq \sum_{i \neq k} \epsilon^{-1} = 1 - P^*$$

and (1) follows.

Remarks:

Sharper inequalities than that used here are available for martingales (Chow (1960) and Robbins (1970), p. 1400, 1401); we will not, however, pursue this possibility here. It is really sufficient in (A) and (B) to have $Z_{ikn}(\tau_{ik})$ a submartingale with $E|Z_{ikn}(\tau_{ik})| = 1$; however, in all our applications we have the stronger requirements (A) and (B) fulfilled. We often find it possible to define $Z_{ijn}(\tau_{ij})$ for all i and j satisfying (A), (B) and (C).

3. Hoel's method

That Hoel's method is a special case of the method above, can be seen as follows: In his notation, let $u(\tau_{ij})$ be any function of τ_{ij} with $u(\tau_o) = \tau_1$ (e.g. we could take $u(\tau_{ij}) = \tau_{ij} - \tau_o + \tau_1$ or $u(\tau_{ij}) = \tau_1 \tau_{ij}/\tau_o$) and put

$$(3) \quad Z_{ijn}(\tau_{ij}) = g_{n,u(\tau_{ij})}(T_{ij}(n))/g_{n,\tau_{ij}}(T_{ij}(n)).$$

The procedure of paragraph 2 above based on the choice $Z_{ijn} = Z_{ijn}(\tau_o)$ clearly is

equivalent to Hoel's procedure. Since $\{Z_{ijn}(\tau_{ij}), n = 1, 2, \dots\}$ forms a non-negative martingale sequence with expectation one, (Doob (1953), p. 93) we get the following alternative to the result of Hoel:

Theorem 2

If $Z_{ikn}(\tau_{ik})$ as defined by (3) is non-increasing in τ_{ik} , and Hoel's procedure stops with probability one, then (1) holds for his procedure.

It would seem that the condition of this theorem can usually be verified more easily than condition (5) of Hoel's theorem. An example to this effect is the problem of selecting the stochastically largest population as discussed in paragraph 4 of Hoel's paper: Taking $u(\tau_{ij}) = \tau_1 \tau_{ij} / \tau_0$ we have here

$$\begin{aligned} Z_{ijn}(\tau_{ij}) &= \left(\frac{\tau_1}{\tau_0}\right)^n \prod_{\nu=1}^n \frac{\Gamma(s_{\nu n}(ij) + \nu(\tau_1 \tau_{ij} / \tau_0 - 1)) \Gamma(s_{\nu+1, n}(ij) + \nu(\tau_{ij} - 1))}{\Gamma(s_{\nu+1, n}(ij) + \nu(\tau_1 \tau_{ij} / \tau_0 - 1)) \Gamma(s_{\nu n}(ij) + \nu(\tau_{ij} - 1))} \\ &= \left(\frac{\tau_1}{\tau_0}\right)^n \prod_{\nu=1}^n J_{\nu n}(\tau_{ij}). \end{aligned}$$

To show that $Z_{ijn}(\tau_{ij})$ is non-increasing in τ_{ij} , it suffices to show that $J_{\nu n}(\tau_{ij})$ is non-increasing in τ_{ij} for each ν . But

$$(4) \quad J_{\nu n}(\tau_{ij}) = \prod_{\alpha=1}^{s_{\nu+1, n}(ij) - s_{\nu n}(ij)} K(s_{\nu+1, n}(ij) - \alpha, \tau_{ij})$$

with

$$K(a, \tau_{ij}) = [a + \nu(\tau_{ij} - 1)] / [a + \nu(\tau_1 \tau_{ij} / \tau_0 - 1)].$$

It is easy to see that $K(a, \tau_{ij})$ is non-increasing for the range of values of a in (4) as long as $\tau_1 / \tau_0 > 1$, which here is assumed to be the case. The result therefore follows and we can conclude that if the procedure stops (see Hoel (1971), p. 638 for discussion on this matter) then (1) would hold.

A more general class of possible choices of the Z 's is this: let $F(\tau_1)$ be a df placing all its mass on $[\tau_0, \infty)$ and put e.g.

$$(5) \quad Z_{ijn}(\tau_{ij}) = \int g_{n, \tau_1 \tau_{ij} / \tau_0}(\mathbf{T}_{ij}(n)) dF(\tau_1) / g_{n, \tau_{ij}}(\mathbf{T}_{ij}(n))$$

(we could replace $\tau_1 \tau_{ij} / \tau_0$ by $\tau_{ij} - \tau_0 + \tau_1$, etc.) Then $\{Z_{ijn}(\tau_{ij}), n=1, 2, \dots\}$ still

forms a non-negative martingale sequence with expectation one and we have to check only (C) and termination of the resulting procedure.

4. Asymptotic expected number of observations

In this section we give a general result which can be used to establish the AENO and AENS as $P^* \rightarrow 1$. The idea behind this result is similar to that of Perng (1969).

Consider a procedure such as that defined in section 2 and suppose that (1) holds for it; let T denote the total number of observations required to stop and N the number of stages. Suppose also that the event $Z_{ijn} \geq \epsilon$ is equivalent to

$$(6) \quad Y_{ijn} \geq \rho(n, P^*)$$

and that the following conditions hold:

(I) For each i and j , $Y_{ijn} \rightarrow m_{ij}$ a.s. as $n \rightarrow \infty$, where m_{ij} are constants such that $\rho_0 < m_{kj} = \max_i m_{ij}$ for each j , with ρ_0 some finite constant.

(II) There exists an integer $n_0 \geq 1$ and a number P_0 with $0 < P_0 < 1$ such that

(i) For each fixed $n \geq n_0$, $\rho(n, P^*)$ is non-decreasing in $P^* \geq P_0$ and $\rho(n, P^*) \uparrow \infty$ as $P^* \uparrow 1$;

(ii) for each fixed $P^* \geq P_0$, $\rho(n, P^*)$ is non-increasing in n for $n \geq n_0$, and $\lim_{n \rightarrow \infty} \rho(n, P^*) \leq \rho_0$.

(III) For each $x > \rho_0$, define $r(x, P^*) = \inf\{n \geq n_0 : \rho(n, P^*) \leq x\}$ and assume that $r(x, P^*) \sim g(P^*)\ell(x)$ as $P^* \rightarrow 1$ with $g(P^*)$ a function of P^* only, $g(P^*) \rightarrow \infty$ as $P^* \rightarrow 1$, and $\ell(x)$ a function of x only continuous in $x > \rho_0$.

(IV) There exists a random variable R (depending on P^*) such that $N \leq R$ a.s. with $R/g(P^*)$ convergent in probability and $\{R/g(P^*)\}$ a uniformly integrable family.

Then we have

Theorem 3

If (I)–(IV) hold, then

$$(7) \quad \lim_{P^* \rightarrow 1} EN/g(P^*) = \max_{j \neq k} \ell(m_{kj})$$

$$(8) \quad \lim_{P^* \rightarrow 1} ET/g(P^*) = \sum_{j \neq k} \ell(m_{kj}) + \max_{j \neq k} \ell(m_{kj}).$$

Proof:

All integers n , n_i etc. to be referred to below will be assumed $\geq n_0$ without further mention; similarly, we consider $P^* \geq P_0$ only.

With (Ω, F, P) denoting the underlying probability space, put

$$\Omega_0 \equiv CS = \{ \omega \in \Omega : \Pi_k \text{ is selected} \}.$$

Then

$$ET = \int_{\Omega_0} T dP + \int_{\Omega_0^c} T dP.$$

Now $P(\Omega_0^c) \leq 1 - P^* \rightarrow 0$ as $P^* \rightarrow 1$; since $T \leq kR$ a.s., it follows from (IV) that $\int_{\Omega_0^c} T/g(P^*)dP \rightarrow 0$; therefore (8) holds if $\int_{\Omega_0} T/g(P^*)dP$ tends to the same limit. Define N_j , $j \neq k$, on Ω_0 as the first integer $n \geq 1$ such that $\max_i Y_{ijn} \geq \rho(n, P^*)$, the max here being taken over those Π 's remaining at stage $n-1$; thus N_j is the number of observations on Π_j required for its elimination. Clearly, on Ω_0 ,

$$(9) \quad T = \sum_{j \neq k} N_j + \max_{j \neq k} N_j.$$

By (I) and Egoroff's theorem, for each $\delta > 0$ there exist disjoint sets A_{ij} and B_{ij} with $A_{ij} \cup B_{ij} = \Omega$, $P(B_{ij}) < \delta$ and $Y_{ijn} \rightarrow m_{ij}$ uniformly in ω for $\omega \in A_{ij}$; thus, there exist integers $n_{ij} = n_{ij}(\delta)$ such that for all $\omega \in A_{ij}$ and for all $n \geq n_{ij}$ we have

$$(10) \quad m_{ij} - \delta \leq Y_{ijn} \leq m_{ij} + \delta.$$

Put $n_1 = \max_{ij} n_{ij}$, $\Omega_1 = \Omega_0 \cap \bigcup_{ij} B_{ij}$ and $\Omega_2 = \Omega_0 - \Omega_1$. Then, from (10), for $n \geq n_1$, on Ω_2 , we have

$$(11) \quad m_{kj} - \delta \leq \max_i Y_{ijn} \leq m_{kj} + \delta$$

where the max is over those Π_j 's remaining at stage $n-1$ and we have $\max_i m_{ij} = m_{kj}$ in view of (I) and the fact that Π_k is never eliminated on Ω_0 . Put $\Omega_3 = \{ \omega \in \Omega_2 : N_j < n_1 \text{ for some } j \}$ and $\Omega_4 = \Omega_2 - \Omega_3$. Then

$$(12) \quad \int_{\Omega_0} N_j dP = \int_{\Omega_1} N_j dP + \int_{\Omega_3} N_j dP + \int_{\Omega_4} N_j dP.$$

Now $P(\Omega_1) \leq k^2 \delta$; hence, from (IV), $\lim_{\delta \rightarrow 0} \limsup_{P^* \rightarrow 1} \int_{\Omega_1} N_j/g(P^*)dP = 0$. It is easy to

see that $N_j \uparrow \infty$ a.s. as $P^* \uparrow 1$; hence $\Omega_3 \downarrow$ as $P^* \uparrow$ and $P(\Omega_3) \downarrow 0$ as $P^* \uparrow 1$; hence, from (IV) $\int_{\Omega_3} N_j/g(P^*)dP \rightarrow 0$ as $P^* \rightarrow 1$. Finally, consider Ω_4 , here $N_j \geq n_1$ and (11) holds; applying the definition of N_j and (III) we get

$$(13) \quad r(m_{kj} + \delta, P^*) \leq N_j \leq r(m_{kj} - \delta, P^*) + 1.$$

Now $P^* - P(\Omega_3) - k^2\delta \leq P(\Omega_4) \leq P^*$; dividing (13) by $g(P^*)$ and using (III), we get

$$\lim_{\delta \rightarrow 0} \limsup_{P^* \rightarrow 1} \int_{\Omega_4} N_j/g(P^*)dP \leq \ell(m_{kj}) \text{ and}$$

$$\lim_{\delta \rightarrow 0} \liminf_{P^* \rightarrow 1} \int_{\Omega_4} N_j/g(P^*)dP \geq \ell(m_{kj});$$

putting these results together in (12), letting $P^* \rightarrow 1$ and then $\delta \rightarrow 0$, we get

$$\int_{\Omega_0} N_j/g(P^*)dP \rightarrow \ell(m_{kj}). \text{ Similarly one shows that } \int_{\Omega_0} \max_{j \neq k} N_j/g(P^*)dP \rightarrow \max_{j \neq k} \ell(m_{kj})$$

and (8) follows. (7) follows analogously.

Remarks:

In the above proof condition (IV) was used to ensure that

$$(IV_1) \int_{\Omega_0^c} N/g(P^*)dP \rightarrow 0 \text{ and } \int_{\Omega_3} N/g(P^*)dP \rightarrow 0 \text{ as } P^* \rightarrow 1 \text{ and that}$$

$$(IV_2) \lim_{\delta \rightarrow 0} \limsup_{P^* \rightarrow 1} \int_{\Omega_1} N/g(P^*)dP = 0.$$

In some applications we find it easier to accomplish this by means other than verification of (IV). Of course, if the procedure is truncated, i.e. $N \leq c$, a finite constant, then (IV) and these conditions hold automatically.

5. Applications

In this section we give some examples of the application of the general results of the preceding paragraphs.

5.1 Selecting the population with largest mean

In this section we suppose that $\mu_i = EX_{i\alpha}$ exists for all i and that the distributions of $X_{i\alpha} - \mu_i$ are identical for all i ; we put $\tau_{ij} = \mu_j - \mu_i$; then the best population is that having largest mean $\mu_k = \max_i \mu_i$. Put $\Psi(b) = E \exp(b(X_{i\alpha} - \mu_i))$ assuming this moment generating function to exist on some set B with $B - \{0\} \neq \emptyset$.

Let F be a df placing all its mass on $\{x \geq 0 : x \in B, -x \in B\}$ but not all on $\{0\}$; then define

$$(14) \quad Z_{ijn}(\tau_{ij}) = H_n(S_{in} - S_{jn} - n\tau_{ij})$$

with

$$(15) \quad H_n(x) = \int_0^\infty \exp(bx) \Psi(b)^{-n} \Psi(-b)^{-n} dF(b)$$

and

$$(16) \quad S_{in} = \sum_{\alpha=1}^n X_{i\alpha},$$

assuming the integral in (15) to be finite for all x and n .

Then $\{Z_{ijn}(\tau_{ij}), n = 1, 2, \dots\}$ forms a non-negative martingale satisfying (A), (B) and (C) of paragraph 2 and hence, if the resulting procedure stops with probability one, then it satisfies (1). Note that $Z_{ijn}(\tau_{ij})$ here may depend on parameters other than τ_{ij} ; as long as $Z_{ijn}(\tau_0) = Z_{ijn}$ does not depend on any unknown parameters an executable procedure results from this construction.

This procedure can be put in the equivalent form: At stage n Π_i eliminates Π_j if

$$(17) \quad Y_{ijn} \geq \tau_0 + n^{-1} H_n^{-1}(\epsilon) = \rho(n, P^*)$$

with

$$Y_{ijn} = n^{-1}(S_{in} - S_{jn}).$$

Now a sufficient condition under which the procedure stops with probability 1 is this: if $E(X_{i\alpha} - \mu_i)^2 = \sigma^2 < \infty$ and if

$$(18) \quad \rho(n, P^*) \leq 2A\sigma(\log \log n/n)^{1/2} \text{ for all } n \text{ sufficiently large,}$$

where $0 < A < 1$, then the procedure stops with probability one. To see this, suppose to the contrary that for some ω the procedure does not stop; then there must be at least two populations Π_i and Π_j , say, which are never eliminated. If $\tau_{ij} \neq 0$ then either $\tau_{ij} > 0$ or $\tau_{ji} = -\tau_{ij} > 0$; if $\tau_{ij} > 0$ then apart from ω 's in a null set, we have $Y_{ijn} \rightarrow \tau_{ij} > 0$ so that for some large enough n we would have $Y_{ijn} \geq \rho(n, P^*)$ and eliminate Π_j — a contradiction. Similarly the case $\tau_{ji} > 0$ yields a contradiction. Finally, for $\tau_{ij} = 0$, by the law of the iterated logarithm, apart from ω in a null set,

we have $Y_{ijn} > 2A\sigma(\log \log n/n)^{1/2} \geq \rho(n, P^*)$ infinitely often so that again a contradiction results.

A sufficient condition for the procedure to be truncated is the somewhat stronger one

$$(19) \quad \rho(n_1, P^*) \leq 0 \text{ for some integer } n_1, \text{ for,}$$

under this condition, we have $\max_i Y_{ijn_1} = \max_i n_1^{-1}(S_{in_1} - S_{jn_1}) \geq 0 \geq \rho(n_1, P^*)$ for all Π_j 's remaining at stage n_1 ; hence all populations would then be eliminated and the procedure truncated at stage n_1 .

Next we consider calculation of the asymptotic expected number of observations and we restrict attention here to the case where the procedure is truncated (so that condition (IV) of Theorem 3 holds). In condition (I) we have $m_{ij} = \tau_{ij} = \mu_i - \mu_j$ so that $m_{kj} = \max_i m_{ij}$ for each j and $m_{kj} > \rho_0 = 0$ as long as Π_k is the unique best population (in particular, if $\mu_k - \mu_j \geq \tau^* > 0$). The first part of (II) follows from the monotonicity of $H_n(x)$ in x and we assume here that F and Ψ is such that the second part of (II) holds. Then we show that (III) holds with

$$(20) \quad g(P^*) = \log \epsilon = \log((k-1)/(1-P^*)), \quad \ell(x) = 1/\log B(x) \text{ where}$$

$$(21) \quad B(x) = \text{a.s. sup of } e^{b(x+\tau^*)}/\Psi(b)\Psi(-b) \text{ w.r.t. } F(b).$$

From the definition of $r = r(x, P^*)$ we find that

$$(22) \quad \int_0^\infty [\exp(b(x+\tau^*))\Psi(b)^{-1}\Psi(-b)^{-1}]^{r-1} dF(b) < \epsilon \leq \int_0^\infty [\exp(b(x+\tau^*))\Psi(b)^{-1} \\ \cdot \Psi(-b)^{-1}]^r dF(b)$$

Now, if for some sequence P_γ^* , $\gamma = 1, 2, \dots$ with $P_\gamma^* \uparrow 1$ we have $r(x, P_\gamma^*) \rightarrow m < \infty$, then, for all γ large enough $r(x, P_\gamma^*) = m$ so that the right-hand side of (22) would be finite, while $\epsilon = (k-1)/(1-P_\gamma^*)$ becomes arbitrarily large as $\gamma \rightarrow \infty$ — a contradiction. Hence $r(x, P^*) \rightarrow \infty$ as $P^* \rightarrow 1$ for each x . Then, using the fact that

$$(\int_0^\infty [\exp(b(x+\tau^*))\Psi(b)^{-1}\Psi(-b)^{-1}]^r dF(b))^{1/r} \rightarrow B(x) \text{ as } r \rightarrow \infty,$$

it follows from (22) that $r(x, P^*)/g(P^*)\ell(x) \rightarrow 1$ as $P^* \rightarrow 1$ and (III) holds. Theorem 3 is then applicable.

5.1.1 Normal case

We now specialize to the case where $X_{i\alpha}$ is $N(\mu_i, \sigma^2)$ distributed with σ^2 known; then $\Psi(b) = \exp(b^2\sigma^2/2)$ and

$$(23) \quad H_n(x) = \int_0^\infty \exp(bx - nb^2\sigma^2) dF(b)$$

which is finite for all n and x . We assume F is such that $\rho(n, P^*)$ is non-increasing in n . The requirement (19) for truncation, is equivalent here to

$$(24) \quad F \text{ places positive mass on } (0, \tau^*/\sigma^2).$$

To see this, suppose (19) holds but (24) does not hold. Now $\rho = \rho(n, P^*)$ is uniquely characterized by the relation

$$\epsilon = \int_0^\infty e^{nb(\rho + \tau^* - b\sigma^2)} dF(b).$$

For n_1 $\rho(n_1, P^*) \leq 0$, by (19); hence

$$\begin{aligned} \epsilon &\leq F(\{0\}) + \int_{[\tau^*/\sigma^2, \infty)} e^{n_1 b(\tau^* - b\sigma^2)} dF(b) \\ &\leq F(\{0\}) + F([\tau^*/\sigma^2, \infty)) \\ &= 1 \end{aligned}$$

which is a contradiction. Conversely, if (24) holds but not (19), then there exists a_1, a_2 with $0 < a_1 < a_2 < \tau^*/\sigma^2$ such that $F([a_1, a_2]) > 0$ while on $[a_1, a_2]$ we have

$$\exp[b(\rho + \tau^* - b\sigma^2)] \geq 1 + a_3 \quad \text{with } a_3 > 0;$$

hence $\epsilon \geq (1 + a_3)^n F([a_1, a_2])$ which yields a contradiction for n large enough.

We are not yet able to characterize (18) in terms of F ; it would also be of interest to characterize the requirement that $\rho(n, P^*)$ be non-increasing in terms of F .

Next, assuming that (24) holds, the procedure is truncated and we can apply the above results to get the AENO. Here

$$B(x) \equiv B_F(x) = \text{a.s. sup } \exp[b(x + \tau^*) - b^2\sigma^2] \text{ w.r.t. } F(b).$$

It is clear that $B_F(x)$ is non-decreasing in x and as $\exp[b(\tau^* - b\sigma^2)] > 1$ for $b \in (0, \tau^*/\sigma^2)$ we have $B_F(0) > 1$, i.e. $B_F(x) > 1$ for all $x > 0$; hence $\ell_F(x) = 1/\log B_F(x) > 0$ for $x > 0$. Further since

$$\exp[b(x + \tau^* - b^2\sigma^2)] \leq \exp[(x + \tau^*)^2/4\sigma^2] \text{ or all } b \geq 0, x > 0,$$

we have

$$B_F(x) \leq \exp[(x + \tau^*)^2/4\sigma^2]$$

and

$$\ell_F(x) \geq 4\sigma^2/(x + \tau^*)^2$$

with equality here if e.g. F places positive mass in all neighbourhoods of every point $x > 0$.

Thus, from the point of view of having smallest AENO as $P^* \rightarrow 1$, the best choice of F would be one having $[0, \infty)$ as support.

It is of interest to note some special choices of F :

- (a) If F places all its mass at the point $b = (\tau_1 - \tau_0)/2\sigma^2$ with $\tau_1 > \tau_0$, then we get Paulson's procedure (Hoel (1971), p. 633); (24) becomes $\tau_1 - \tau_0 < 2\tau^*$ i.e. $\tau_1 < \tau^* = -\tau_0$ which is the familiar condition under which Paulson's procedure is truncated. It also stops a.s. if $\tau_1 \leq \tau^*$ as can be seen from (18) and that it satisfies (1) follows from our general result. Further,

$$\ell(x) = 1/[b(x + \tau^*) - b^2\sigma^2] \text{ with } b = (\tau_1 - \tau_0)/2\sigma^2$$

and the resulting expression for the AENO agrees with that of Perng (1969).

- (b) If $dF(b) = (2/\pi)^{1/2} \exp(-b^2/2)db$, $b \geq 0$ then

$$H_n(x) = 2\exp(\frac{1}{2}x^2/(1+2n\sigma^2))\Phi(y/(1+2n\sigma^2)^{1/2})(1+2n\sigma^2)^{-1/2}$$

with Φ the unit normal df. An explicit expression for $\rho(n, P^*)$ is not possible, but fairly sharp bounds are

$$\begin{aligned} \tau_0 + n^{-1}(1 + 2n\sigma^2)^{1/2}[2 \log \epsilon/2 + \log(1 + 2n\sigma^2)]^{1/2} \\ \leq \rho(n, P^*) \leq \tau_0 + n^{-1}(1 + 2n\sigma^2)^{1/2}[2 \log \epsilon + \log(1 + 2n\sigma^2)]^{1/2} \end{aligned}$$

If we replace the procedure by one in which $\rho(n, P^*)$ is taken equal to the upper bound above, (1) would still hold, the procedure is still truncated and has the same AENO, viz. that of (7) with $\ell(x) = 4\sigma^2/(x+\tau^*)^2$. This procedure is that discussed in Swanepoel and Geertsema (1972) where it was noted that it has a better asymptotic performance as $P^* \rightarrow 1$ than the procedure of Paulson.

It is perhaps worth remarking that if the distribution of Π_i is not necessarily normal but such that $\Psi(b) \leq \exp(b^2\sigma^2/2)$ for all b , then we can still apply this last procedure; it will satisfy (1) and has the same AENO.

Finally we remark that the class of procedures discussed above can also be derived from integrated likelihood ratio martingales as in (5).

5.2 Selecting the normal population with smallest variance

Next we consider the case where $X_{i\alpha}$ is $N(\mu_i, \sigma_i^2)$ distributed with both μ_i and σ_i^2 's unknown. Two possible definitions of τ_{ij} will be treated:

5.2.1 The choice $\tau_{ij} = \sigma_j^2/\sigma_i^2$

This case is treated by Hoel (1971), p. 634; we discuss the more general procedure mentioned in (5) above. Following Hoel's notation, we define

$$Z_{ijn}(\tau_{ij}) = \int_{\tau_0}^{\infty} \left[\frac{1/\tau_{ij} + T_{ij}(n)}{\tau_0/\tau_1\tau_{ij} + T_{ij}(n)} \left(\frac{\tau_0}{\tau_1} \right)^{1/2} \right]^{n-1} dF_1(\tau_1)$$

with F_1 placing all its mass on $[\tau_0, \infty)$. This is easily shown to be non-increasing in τ_{ij} and hence, provided that the resulting procedure stops, it satisfies (1).

Making the substitution $b = \tau_1/\tau_0$, we get

$$(25) \quad Z_{ijn}(\tau_{ij}) = H_n(\tau_{ij}T_{ij}(n))$$

where

$$(26) \quad H_n(x) = \int_1^{\infty} ((1+x)b^{1/2}/(1+bx))^{n-1} dF(x)$$

with F placing all its mass on $[1, \infty)$. We assume now that F does not place all its mass on $\{1\}$; then $H_n(x)$ is strictly decreasing in x ; hence $Z_{ijn}(\tau_{ij})$ is non-increasing in τ_{ij} and satisfies (C). The actual procedure is based on $Z_{ijn} = Z_{ijn}(\tau_0)$ and can essentially be reformulated as: Π_i eliminates Π_j if

$$(27) \quad Y_{ijn} = T_{ji}(n) \geq \tau_0/H_n^{-1}(\epsilon) = \rho(n, P^*).$$

A sufficient condition for truncation in this case is

$$(28) \text{ for some } n_1 = n_1(P^*), \rho(n_1, P^*) \leq 1,$$

for then $\max_i Y_{ij} n_1 \geq 1 \geq \rho(n_1, P^*)$ so that all populations would be eliminated at this stage. This condition is equivalent to the requirement

$$(29) \text{ F places positive mass on } (1, \tau^{*2}).$$

This follows by an argument analogous to that given in paragraph 5.1.1. To get an expression for the AENO we can again apply Theorem 3; in this case $m_{ij} = \sigma_i^2/\sigma_1^2 = \tau_{ij}$; again $g(P^*) = \log e$ and $\ell(x) \equiv \ell_F(x) = 1/\log B_F(x)$ where now

$$(30) \text{ } B_F(x) = \text{a.s. sup of } (x + \tau_0)b^{1/2}/(x + b\tau_0) \text{ w.r.t. } F(b).$$

Now

$$(x + \tau_0)b^{1/2}/(x + b\tau_0) \leq ((x/\tau_0)^{1/2} + (\tau_0/x)^{1/2})/2$$

for all $b \geq 1$ and $x > \tau_0$; hence

$$B_F(x) \leq ((x/\tau_0)^{1/2} + (\tau_0/x)^{1/2})/2, \ell_F(x) \geq 1/\log[((x/\tau_0)^{1/2} + (\tau_0/x)^{1/2})/2]$$

with equality if e.g. F places positive probability on every neighbourhood of every point of $[1, \infty)$. Such a choice therefore yields the best procedure from an AENO point of view.

The procedure discussed by Hoel corresponds to the special choice where F places all its mass on $b = \tau_1/\tau_0$ with $\tau_0 < \tau_1$; for truncation (29) requires that $\tau_1/\tau_0 < \tau^{*2}$ i.e. $\tau_1 < \tau^*$ which is the same as required by Hoel. The AENO in Hoel's case is given by theorem 3 with

$$\ell(x) = 1/\log[(x + \tau_0)(\tau_1/\tau_0)^{1/2}/(x + \tau_1)].$$

5.2.2 The choice $\tau_{ij} = 1/\sigma_i^2 - 1/\sigma_j^2$

Hoel (1971), p. 637 mentions that a procedure with this choice of τ_{ij} can be constructed along the lines of Paulson's original derivation. This procedure can also be obtained as a special case of our result of paragraph 2; we briefly indicate this.

Define

$$Z_{ijn} = \exp[\frac{1}{2}\tau^*(n-1)(s_{jn}^2 - s_{in}^2)]$$

with the s_{in} 's as above and τ^* the indifference constant for the present case. The procedure obtained from this Z_{ijn} is precisely that mentioned by Hoel. To see how Theorem 1 applies, define.

$$Z_{ikn}(\tau_{ik}) = e^{\frac{1}{2}\tau^*(n-1)(s_{kn}^2 - s_{in}^2)} / E e^{\frac{1}{2}\tau^*(n-1)(s_{kn}^2 - s_{in}^2)}$$

Since outside the indifference zone $\tau^* < 1/\sigma_k^2$, the moment generating function appearing here, is finite and it is easy to see that $Z_{ikn}(\tau_{ik})$ forms a non-negative martingale with expectation one. On simplifying, we find that

$$Z_{ikn}(\tau_{ik}) = (1 - \tau^*(\tau_{ik} + \tau^*)\sigma_i^2\sigma_k^2)^{\frac{n-1}{2}} Z_{ikn}$$

which equals Z_{ikn} for $\tau_{ik} = \tau_0 = -\tau^*$ and is non-increasing in τ_{ik} for $\tau_{ik} \leq \tau_0 = -\tau^*$; hence (C) holds. That the procedure stops a.s. can be seen from the law of the iterated logarithm and it therefore follows that it satisfies (1).

In order to apply Theorem 3, we write the procedure in the form: Π_i eliminates Π_j if $s_{jn}^2 - s_{in}^2 \geq 2 \log \epsilon / \tau^*(n-1) = \rho(n, P^*)$. Apart from (IV), the conditions of Theorem 3 clearly hold with $m_{ij} = \sigma_j^2 - \sigma_i^2$, $g(P^*) = \log \epsilon$ and $\ell(x) = 2/\tau^*x$. Since the procedure is not truncated (IV) (or its uses in the proof, i.e. (IV₁) and (IV₂)) is more difficult in this case. It is tedious but possible to show that (IV₁) and (IV₂) hold in the following manner: Define R_{ij} as the smallest n such that $s_{jn}^2 - s_{in}^2 \geq 2 \log \epsilon / \tau^*(n-1)$; then on Ω_0 define $R = \max_{j \neq k} R_{kj}$ and on Ω_0^c define $R = \max_{i \neq j} R_{ij}$. Then we have $N \leq R$, $T \leq kR$ and by representing s_{in}^2 in terms of sums of independent χ_1^2 -variables and using standard sequential results we can verify (IV₁) and (IV₂).

In conclusion we remark that one might have hoped to define a class of procedures e.g. by taking

$$Z_{ijn}(\tau_{ij}) = \int e^{b(n-1)(s_{jn}^2 - s_{in}^2)} dF(b) / E e^{b(n-1)(s_{jn}^2 - s_{in}^2)};$$

however, then $Z_{ijn}(\tau_0)$ is free of unknown parameters only for the choice F degenerate in $b = \frac{1}{2}\tau^*$ considered above.

5.3 Some further applications

5.3.1 Koopman-Darmois families

We remark that it is possible to formulate the procedure of Hoel and Mazumdar

(1968) as a special case of Theorem 1 along the lines of paragraph 5.2.2 above.

The method can however also be used for cases other than the Koopman-Darmois families as is shown by the next application.

5.3.2 Selecting the exponential population with smallest scale parameter

Here we suppose that $X_{i\alpha}$ has density $\sigma_i^{-1} \exp[-(x-\mu_i)/\sigma_i]$ for $x \geq \mu_i$ with both μ_i and σ_i unknown. The construction of a procedure for selecting the smallest variance (without elimination) is discussed by Bechhofer Kiefer and Sobel (1968), p. 118. Here we give an elimination procedure.

Let $Y_{i1n} < Y_{i2n} < \dots < Y_{inn}$ denote the order statistics corresponding to $X_{i1}, \dots, X_{in}$:

let

$$s_{in} = \sum_{\alpha=2}^n (Y_{i\alpha n} - Y_{i1n}).$$

It is not difficult to show that, for each i , if $b\sigma_i < 1$, then

$$\{\exp(b s_{in})(1-b\sigma_i)^{n-1}, n = 1, 2, \dots\}$$

forms a martingale sequence with expectation one.

With $\tau_{ij} = 1/\sigma_i - 1/\sigma_j$ and

$$\begin{aligned} Z_{ikn}(\tau_{ik}) &= \exp[\tau^*(s_{kn} - s_{in})][(1 + \tau^*\sigma_i)(1 - \tau^*\sigma_k)]^{n-1} \\ &= [1 - \tau^*(\tau_{ik} + \tau^*)\sigma_i\sigma_k]^{n-1} Z_{ikn} \end{aligned}$$

we have a procedure entirely analogous to that discussed in 5.2.2 above.

6. Acknowledgement

The authors would like to express their appreciation towards Dr J.C. Geertsema for his interest in this study.

REFERENCES

- BECHHOFFER, R.E., KIEFER, J. and SOBEL, M. (1968). *Sequential identification and ranking procedures*. University of Chicago Press, Chicago.
- CHOW, Y.S. (1960). A martingale inequality and the law of large numbers. *Proc. Amer. Math. Soc.* **11**, 107-111.
- DOOB, J.L. (1953). *Stochastic processes*. J. Wiley & Sons. New York.

- HOEL, D.G. (1971). A method for the construction of sequential selection procedures. *Ann. Math. Statist.*, **42**, 630-642.
- HOEL, D.G. and MAZUMDAR, M. (1968). An extension of Paulson's selection procedure. *Ann. Math. Statist.*, **39**, 2067-2074.
- PERNG, S.K. (1969). A comparison of the asymptotic expected sample sizes of two sequential procedures for ranking problems. *Ann. Math. Statist.* **40**, 2198-2202.
- ROBBINS, H. (1970). Statistical methods related to the law of the iterated logarithm. *Ann. Math. Statist.*, **41**, 1397-1409.
- SWANEPOEL, J.W.H. and GEERTSEMA, J.C. (1972). Sequential procedures with elimination for selecting the best of k normal populations. *Unpublished manuscript*.

Manuscript received 17 August 1974.